

■ NetSage AI — Executive Project Summary

AI-Assisted Cisco Network Troubleshooting & Responsible Governance System

Target Audience: Technical Leads & Network Engineers	System Version: 1.0.0 (Production Ready)
AI Engine: Groq LLaMA 3.3 70B	Verification Status: 62 / 62 Tests Passed (100%)

1. Executive Overview

NetSage AI is an enterprise-grade, evidence-driven Cisco network troubleshooting platform. It bridges the gap between deterministic rule-based network diagnostic verification and advanced LLM natural language reasoning, while enforcing strict **Human-in-the-Loop (HITL) Governance**.

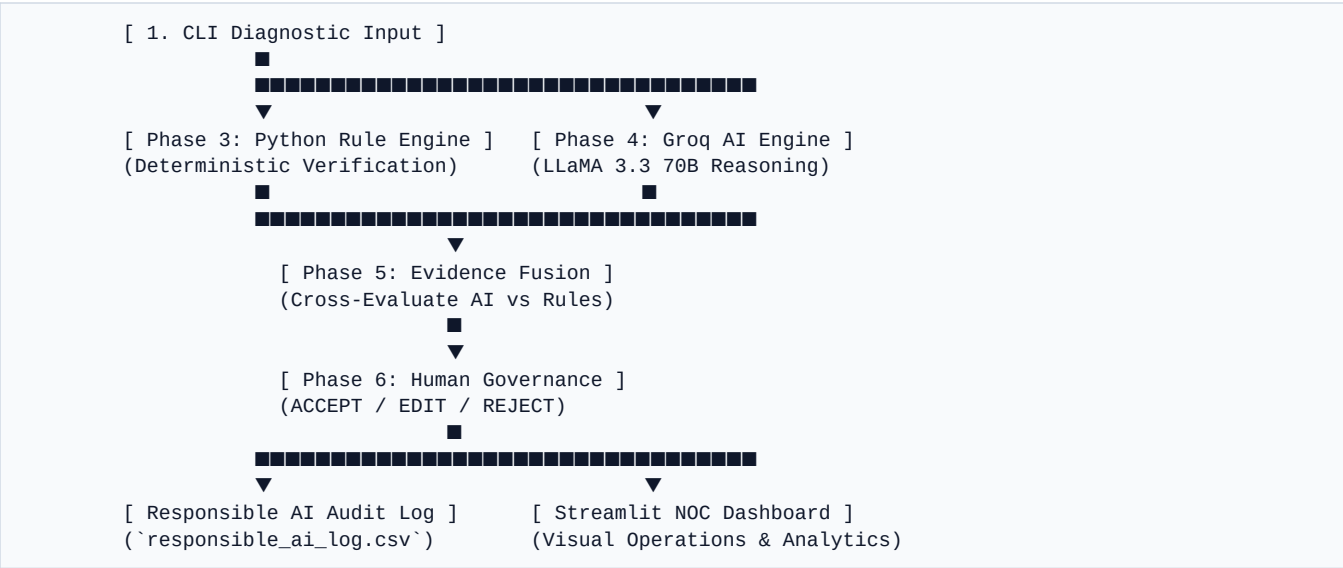
***Governance Philosophy:** Traditional network troubleshooting relying purely on manual CLI analysis is slow and error-prone. Conversely, unguided LLMs introduce severe operational risks such as hallucinations, invalid Cisco IOS syntax, or unauthorized network modifications. NetSage AI resolves this by placing a deterministic verification engine alongside **Groq LLaMA 3.3 70B**, synthesizing their outputs through an **Evidence Fusion Engine**, and requiring **Human Authority** approval before any recommendation is finalized.*

2. Core Metrics & System Specifications

Metric / Specification	Value / Description
Troubleshooting Dataset	40 Lab Cases spanning 8 fundamental Cisco networking domains
Deterministic Rules Engine	6 Python Rules (IF-001, IP-001, IP-002, GW-001, VLAN-001, ROUTE-001)
AI Diagnosis Engine	Groq LLaMA 3.3 70B (llama-3.3-70b-versatile) with Pydantic JSON schema
Evidence Fusion Engine	4 Agreement Categories (AGREE, PARTIAL_AGREE, CONFLICT, INSUFFICIENT_EVIDENCE)
Governance Principles	Human Authority Principle enforcing ACCEPT (85%), EDIT (12.5%), REJECT (2.5%)
Responsible AI Audit Log	6 Genuine Human Corrections (CASE-006, CASE-010, CASE-016, CASE-020, CASE-027, CASE-033)
NOC Operations Dashboard	5 Interactive Pages (Overview, Case Explorer, AI vs Human, Responsible AI, Evaluation)
Automated Test Coverage	62 Unit & Integration Tests Passed (100% Pass Rate in < 0.3s)

3. System Architecture & Workflow Pipeline

NetSage AI processes diagnostic requests through a sequential, non-bypassable 5-stage pipeline:



4. Directory Structure & Key Modules

Module / Path	Responsibility & Key Artifacts
ai/	Groq API client, prompt templates, and Pydantic schemas (diagnosis.py, schemas.py)
rules/	Deterministic rule engine suite evaluating physical, IP, gateway, and VLAN integrity
integration/	Evidence Fusion Engine cross-referencing AI outputs against rules (evidence_fusion.py)
review/	Human Authority governance engine processing ACCEPT/EDIT/REJECT actions and JSONL store
dashboard/	Streamlit NOC application pages, custom glassmorphism styles, and Plotly visualizers
evaluation/	Metrics harness calculating schema validity, precision, recall, and agreement rates
data/	40 troubleshooting cases dataset, pre-computed JSONL runs, and responsible_ai_log.csv
tests/	Full 62 automated unit & end-to-end integration tests (100% pass rate)

5. Automated Test Suite & Quality Verification

Test Module	Coverage Area	Tests	Status
tests/test_rules.py	Deterministic Rules (IF-001 to ROUTE-001)	22	PASS
tests/test_ai_engine.py	Groq AI Engine, JSON schemas, offline fallback	9	PASS
tests/test_integration.py	Evidence Fusion & agreement scoring	6	PASS
tests/test_review.py	Human Review decisions & CSV audit log	8	PASS
tests/test_dashboard_metrics.py	Dashboard analytics & safety limits	9	PASS
tests/test_end_to_end.py	End-to-End troubleshooting pipeline	8	PASS

6. Featured Demo Scenario Walkthrough (CASE-033)

Scenario: Extended Access Control List (ACL) Sequence Order Conflict.

Symptom: Workstation PC-3 (192.168.30.15/24) cannot reach Web Server (10.0.0.5:80). Gateway ping succeeds.

Deterministic Verification: IF-001 PASS, GW-001 PASS (Physical link and gateway reachable).

Groq AI Diagnosis: AI identifies line 10 broad deny ip rule catching traffic before line 20 permit tcp 192.168.30.0/24 host 10.0.0.5 eq 80.

Evidence Fusion: Categorized as AGREE with rule results.

Human AI Governance: Reviewer executes EDIT to clarify top-to-bottom IOS first-match mechanics and logs override to responsible_ai_log.csv.

7. Quick Start & Execution Instructions

```
# 1. Clone repository and setup environment
git clone https://github.com/Omkar-narsale/NetSage-AI.git
cd NetSage-AI
python -m venv venv
.\venv\Scripts\activate

# 2. Install dependencies & run tests
pip install -r requirements.txt
python -m unittest discover -s tests -p "test_*.py"

# 3. Launch Streamlit Operations Dashboard
streamlit run app.py
```